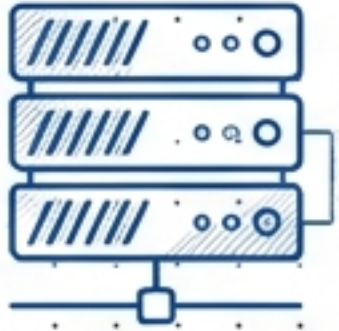


AWS Route 53: Architecture & Mechanics

DNS as a Service – Technical Notes

The As-a-Service Mental Model



EC2 = Compute as a Service



EKS = Kubernetes as a Service



Route 53 = DNS as a Service

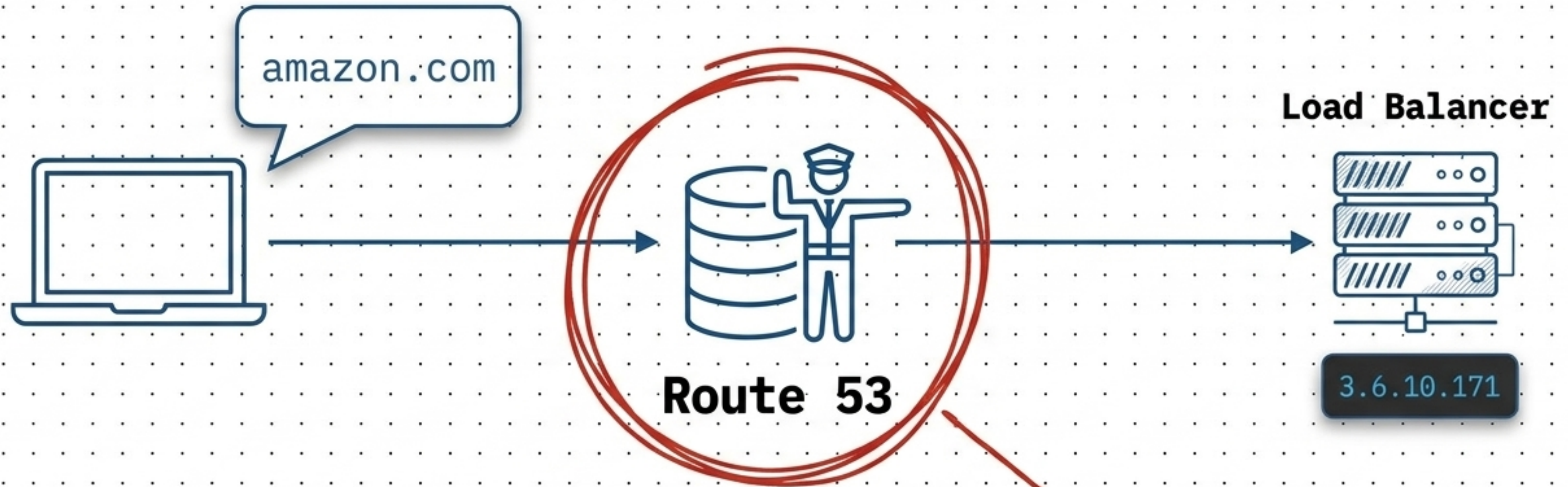
Definition:
The Domain Name System (DNS) is the foundational mapping layer of the internet.

Human vs. Machine Routing Constraints

MACHINES / IP ADDRESSES	HUMANS / DOMAIN NAMES
Example: 3.6.10.171 • Format: Numeric • Nature: Dynamic (changes on reboot or shift) • Memorability: Zero	Example: amazon.com • Format: Semantic string • Nature: Static (permanent anchor) • Memorability: High

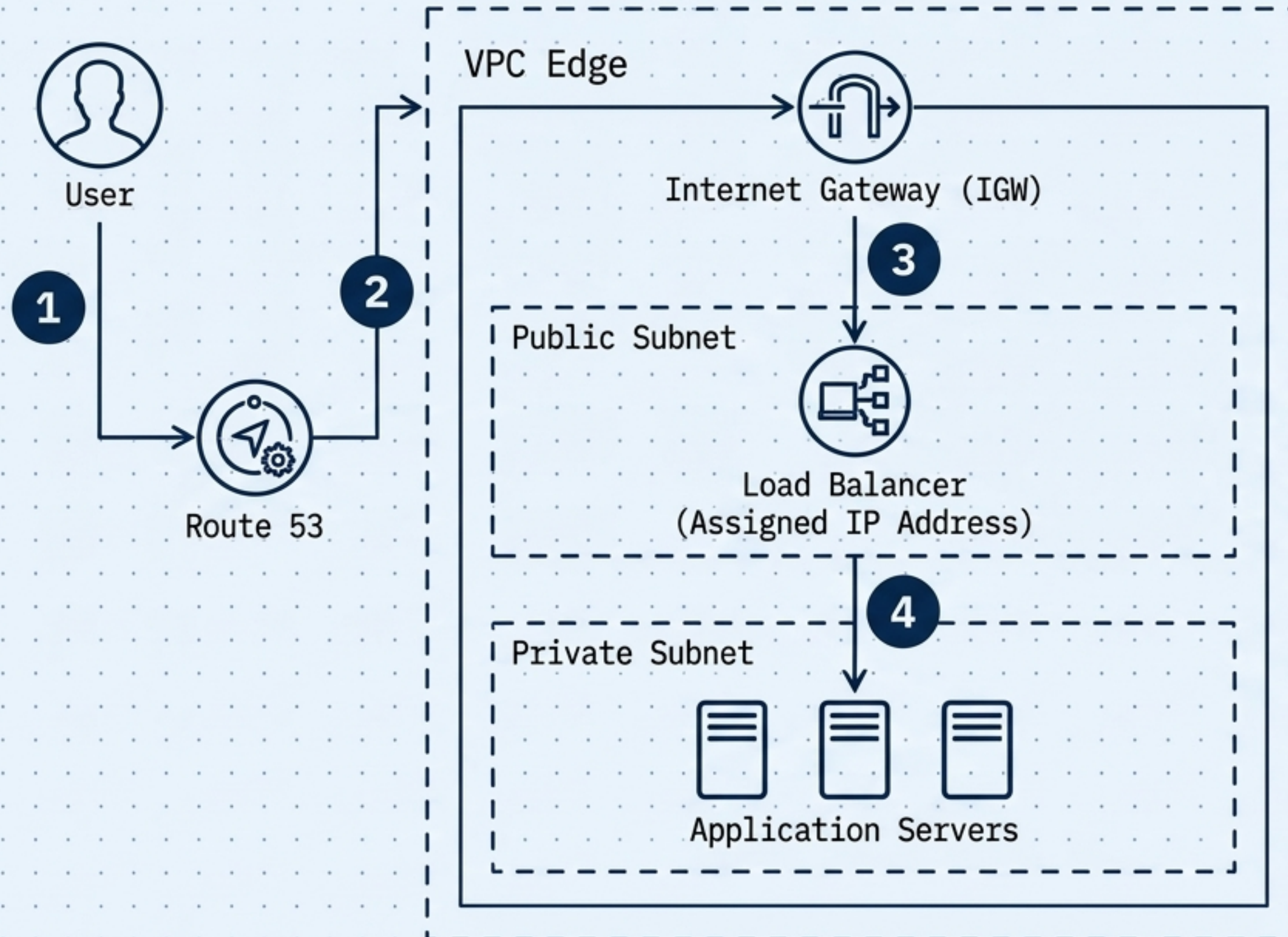
Without a translation layer,
sharing public applications
is practically impossible.

Route 53 at a Glance: The Interceptor



Resolves the name to the precise target IP.

Architectural Context: Mapping to the VPC



LEGEND

- 1** User initiates request to semantic domain name via Route 53
- 2** Traffic passes through the Internet Gateway at the VPC edge
- 3** Request hits the Load Balancer in the Public Subnet holding the resolved IP
- 4** Load Balancer securely forwards traffic to Application Servers in Private Subnet

The Three Pillars of Route 53



Domain Registration

Acquiring the namespace.



Hosted Zones

Storing the mapping records.



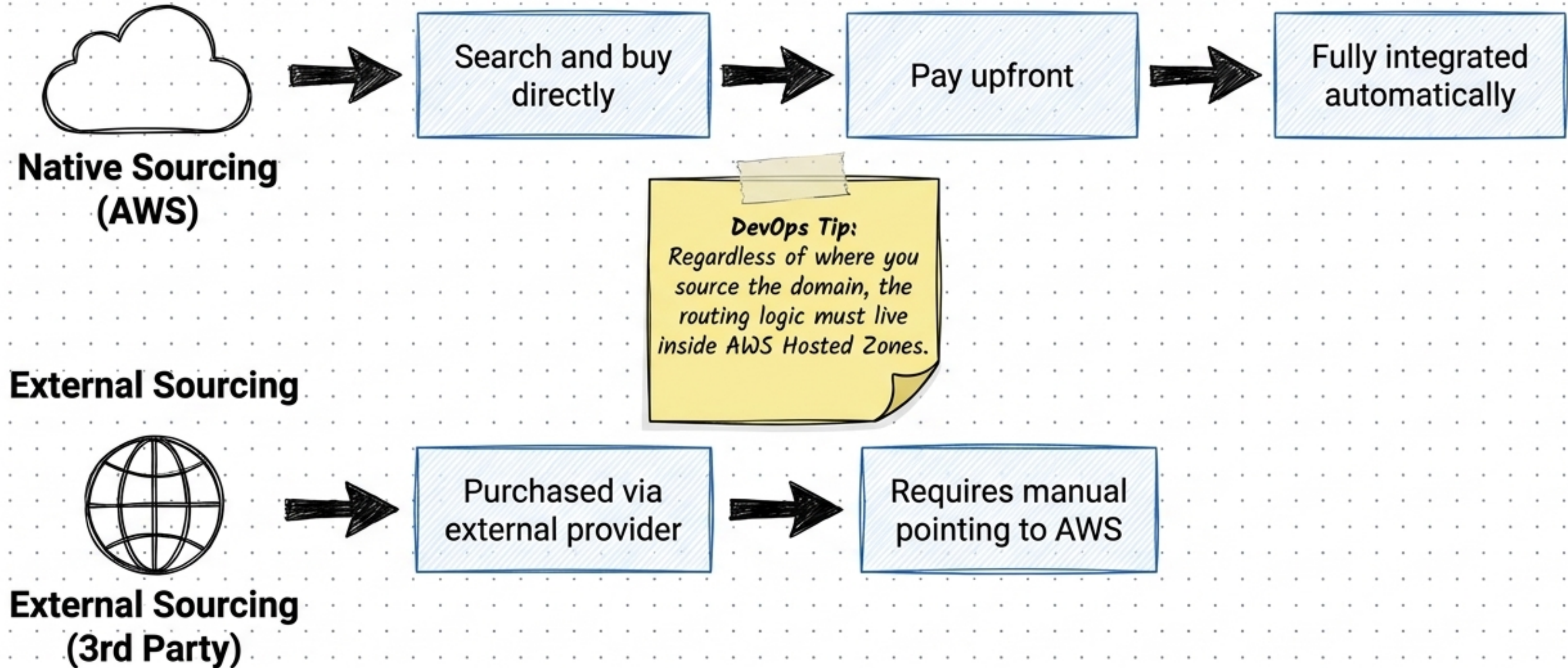
Health Checks

Monitoring target availability.

All three pillars are managed within the Route 53 service console.

Pillar 1: Domain Sourcing & Registration

Split-path workflow

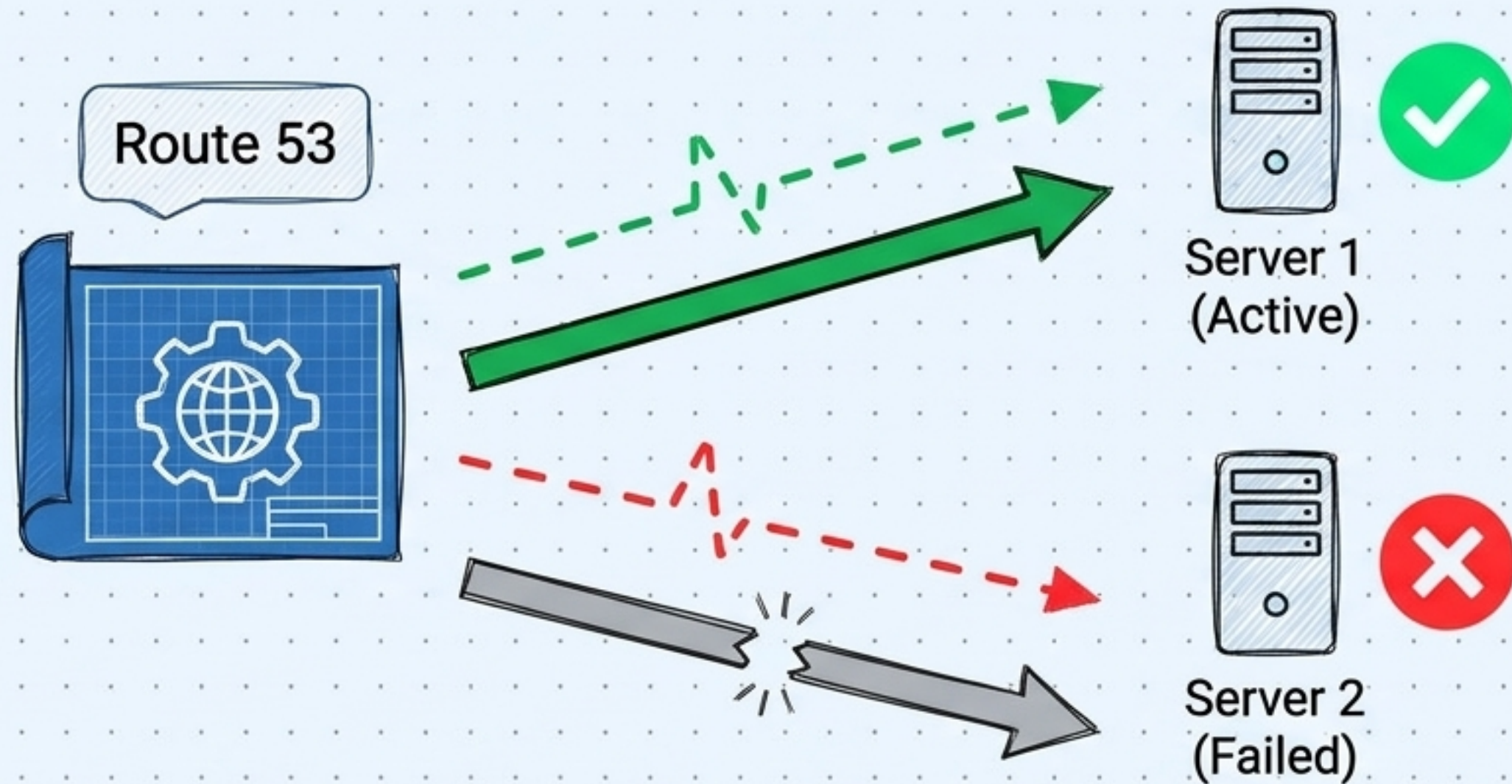


Pillar 2: Hosted Zones & DNS Records

The Core Translation Engine. This is where Route 53 looks upon receiving a user request to determine exactly where to route the traffic.

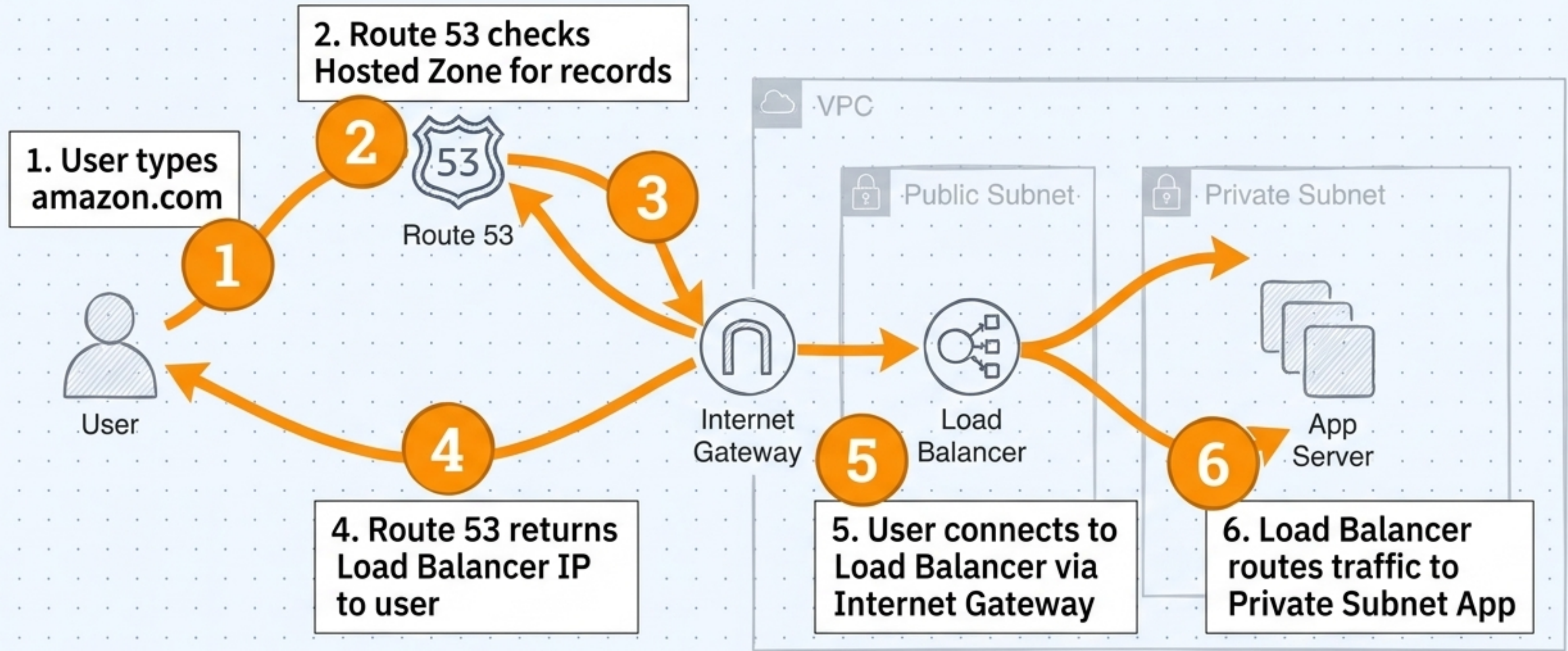
Domain Name (Record)		Target Route (Value)
amazon.com	➡	Load Balancer IP (3.6.10.171)
api.amazon.com	➡	Application IP

Pillar 3: Active Health Checks



Prevents users from experiencing dead links. If a server crashes, Route 53 instantly updates routing to bypass the failed target.

Synthesis: The Complete Request Lifecycle



Implementation Prep & Next Steps

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Practical Lab

Implement the public/private subnet architecture using VPC components and Route 53 integration.

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Interview Prep

Be prepared to whiteboard VPC traffic flow, public/private subnet isolation, and how end applications receive requests.

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Advanced Reading

Review official AWS Route 53 documentation for advanced routing policies (e.g., weighted, latency-based) not covered in the baseline setup.

[EOF]